

Distributions of random variables

Laboratory 2 (old), DEDP

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1 Objective

Generate uniform and normal samples in Matlab for different a , b , μ and σ . Compare histograms and sample statistics with theory, and compute probabilities from data and from CDFs. Use `erf()` for the normal CDF.

2 Theoretical aspects

2.1 Probability density function

In this lab we consider **continuous random variables**.

A continuous random variable A is described by a **probability density function** (PDF) $w_A(x)$, which indicates how likely is that A takes a value x . The PDF is also known as the **distribution** of A .

The probability of A being in the interval $[a, b]$ is the **area under the PDF** in that interval, i.e. the integral of the PDF from a to b :

$$P(a < A \leq b) = \int_a^b w_A(x) dx,$$

Only intervals have nonzero probability. The probability of A being exactly equal to a certain value is always zero.

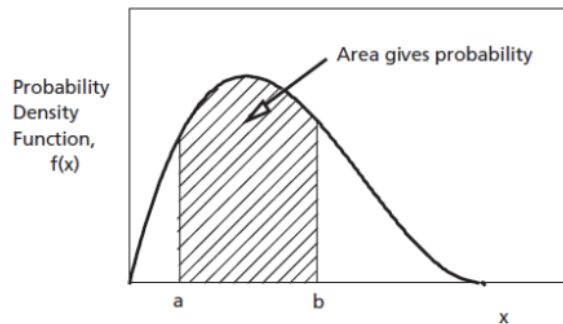


Figure 1: Probability = Area under the PDF

The PDF is always $w_A(x) \geq 0$ and its integral is 1, but its values can be greater than 1.

The **cumulative distribution function** (CDF) is defined as the probability that $A \leq x$:

$$F_A(x) = P(A \leq x)$$

The CDF is nondecreasing and takes values between 0 and 1.

The CDF can also be used to compute probabilities over intervals:

$$P(a < A \leq b) = F_A(b) - F_A(a).$$

The two functions are related. The PDF is the derivative of the CDF:

$$w_A(x) = \frac{dF_A(x)}{dx} \qquad F_A(x) = \int_{-\infty}^x w_A(t) dt.$$

Random samples

One value, or a vector of N generated values from a certain , is a **random sample**:

$$x = [x_1, x_2, \dots, x_N].$$

Values change at every run, but their statistical behavior should remain the same.

The mean and the variance (or standard deviation) computed from a given vector of samples are known as **sample mean** and **sample variance** (or standard deviation).

```
% Sample mean, standard deviation and variance from data
sample_mean = mean(x_normal)
sample_std = std(x_normal)
sample_var = var(x_normal)
```

The sample mean and sample variance are close to, but not exactly equal to, the theoretical mean and variance of the distribution (they are “estimates” of these theoretical values).

2.2 The uniform distribution

Definition and parameters

The uniform distribution describes a random variable that has equal chances of taking any value in a given interval $[a, b]$.

It is denoted as:

$$A \sim \mathcal{U}[a, b], \qquad \text{with } a < b.$$

Its probability density function (PDF) is defined as:

$$f_A(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b, \\ 0, & \text{otherwise.} \end{cases}$$

The PDF is constant on $[a, b]$.

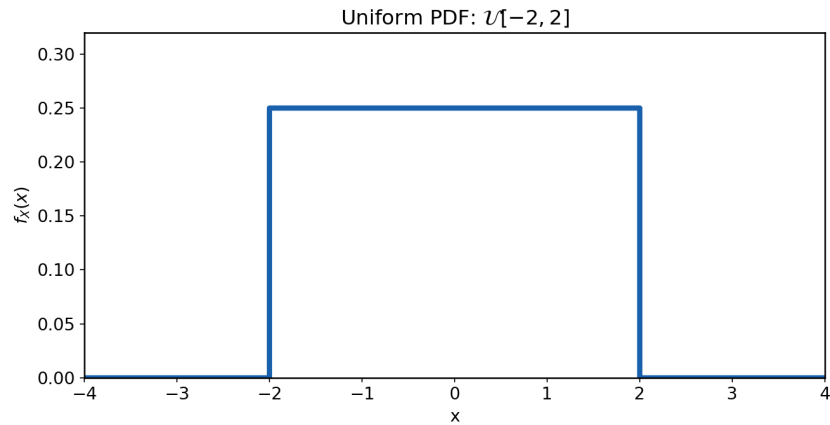


Figure 2: PDF for $\mathcal{U}[-2, 2]$.

Its statistical mean and variance are

$$E\{A\} = \frac{a+b}{2}, \quad \sigma_A^2 = \frac{(b-a)^2}{12}.$$

Generating uniform samples in Matlab

`rand()` generates uniform values with $0 < A < 1$. It does not matter if the endpoints 0 and 1 are included or not, because there are infinitely many values in the interval, anyway.

```
A = rand()           % one value from U[0,1]
A = rand(1, 10)      % row vector with 10 independent values from U[0,1]
```

To obtain values from a general interval $[a, b]$, scale and shift the values:

$$B = a + (b - a)A, \quad A \sim \mathcal{U}[0, 1].$$

```
a = 5;
b = 7;
N = 1000;
x_uniform = a + (b-a)*rand(1, N);
```

Here, $b - a$ sets the width and a sets the lower limit.

Probabilities for the uniform distribution

Probabilities are areas under the PDF.

For a uniform variable, computing probabilities amounts to computing some rectangle area, and can be computed easily.

Example: for $X \sim \mathcal{U}[2, 10]$, the probability that X is between 3 and 5 is $\frac{2}{8} = 0.25$.

2.3 The normal distribution

Definition and parameters

The normal distribution describes a random variable that is more likely to take values near its mean μ than far from it, according to a bell-shaped PDF (“Gaussian hat”).

A normal variable is written as:

$$X \sim \mathcal{N}(\mu, \sigma^2), \quad \sigma > 0.$$

Its probability density function is

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x - \mu)^2}{2\sigma^2}\right].$$

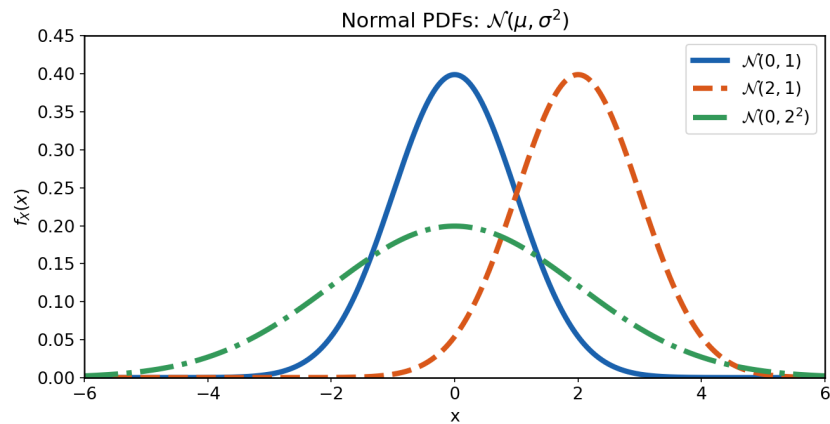


Figure 3: Changing μ shifts a normal PDF; increasing σ makes it wider and flatter.

The parameters:

- μ : mean (the center);
- σ^2 : variance;
- $\sigma = \sqrt{\sigma^2}$: standard deviation.

Changing μ shifts the PDF. Larger σ means the function is wider and flatter, smaller σ means narrower and taller.

The **standard normal distribution** is the normal distribution with $\mu = 0$ and $\sigma = 1$:

$$Z \sim \mathcal{N}(\mu = 0, \sigma^2 = 1).$$

Generating normal samples in Matlab

`randn()` generates values from the standard normal distribution $\mathcal{N}(\mu = 0, \sigma^2 = 1)$:

```
z = randn()           % one value from N(0,1)
z = randn(1, 10)      % 10 independent values from N(0,1)
```

For arbitrary μ and σ , scale with σ and shift with μ :

$$B = \mu + \sigma A, \quad \text{where } A \sim \mathcal{N}(\mu = 0, \sigma^2 = 1).$$

```
mu = 5;
sigma = 2;
N = 1000;
x_normal = mu + sigma*randn(1, N);
```

Note: multiply by the standard deviation σ , not the variance σ^2 .

Computing probabilities from the normal distribution

The integral of the normal PDF does not have a closed form, so it cannot be computed analytically. Instead, we rely on the precomputed `erf()` function in Matlab to compute probabilities.

For a normal variable $A \sim \mathcal{N}(\mu, \sigma^2)$, the probability that A is in the interval $[a, b]$ is:

$$P(a \leq A \leq b) = F_A(b) - F_A(a),$$

where $F_A(x)$ is the cumulative distribution function (CDF) of A and is computed using the error function `erf()`:

$$F_A(x) = P(A \leq x) = \frac{1}{2} \left[1 + \operatorname{erf} \left(\frac{x - \mu}{\sigma \sqrt{2}} \right) \right].$$

In Matlab:

Example: $A \sim \mathcal{N}(\mu = 10, \sigma^2 = 4)$ and $P(8 \leq X \leq 12)$:

```
mu = 10;
sigma = 2;
a = 8;
b = 12;

F_a = 0.5*(1 + erf((a-mu)/(sigma*sqrt(2))));
F_b = 0.5*(1 + erf((b-mu)/(sigma*sqrt(2))));
p_normal_theory = F_b - F_a
```

The result is approximately 0.6827. We can also estimate it from samples as before:

```
N = 10000;
x_normal = mu + sigma*randn(1, N);
p_normal_estimated = sum(x_normal >= a & x_normal <= b)/N;
```

2.4 Histograms as a visualization of distributions

A **histogram** is a bar graph that displays the distribution of samples by counting them in intervals called **bins**:

```
% Generate 10000 samples from a uniform distribution on [-4, 10]
x_uniform = -4 + 14*rand(1, 10000);

% Show a histogram with 30 bins
histogram(x_uniform, 30)
xlabel('Value')
ylabel('Number of samples')
title('Uniform samples')
grid on
```

To get a PDF comparison, we need to normalize its area to 1, using the 'Normalization', 'pdf' option:

```
% Generate 10000 samples from a normal distribution with mean 5 and standard deviation 2
x_normal = 5 + 2*randn(1, 10000);

% Show a PDF-normalized histogram with 35 bins
histogram(x_normal, 35, 'Normalization', 'pdf')
xlabel('Value')
ylabel('Estimated probability density')
title('Normal samples')
grid on
```

Histograms are a practical way to visualize the distribution of samples (values, pixels etc) in order to understand your data.

2.5 Summary: functions and notation you should know

Signature or notation	Example	Description
$X \sim \mathcal{U}[a, b]$	$X \sim \mathcal{U}[-4, 10]$	Uniform distribution on $[a, b]$
$X \sim \mathcal{N}(\mu, \sigma^2)$	$X \sim \mathcal{N}(5, 2^2)$	Normal distribution: mean μ , variance σ^2
rand(m,n)	rand(1,1000)	$m \times n$ samples from $\mathcal{U}[0, 1]$
a + (b-a)*rand(m,n)	-4 + 14*rand(1,1000)	Samples from $\mathcal{U}[a, b]$
randn(m,n)	randn(1,1000)	$m \times n$ samples from $\mathcal{N}(0, 1)$
mu + sigma*randn(m,n)	5 + 2*randn(1,1000)	Samples from $\mathcal{N}(\mu, \sigma^2)$
mean(x), std(x), var(x)	mean(x), std(x), var(x)	Sample mean, standard deviation and variance

Signature or notation	Example	Description
<code>histogram(x,n)</code>	<code>histogram(x,30)</code>	Histogram with <code>n</code> bins; add 'Normalization','pdf' for unit area
<code>erf(x)</code>	<code>erf(1/sqrt(2))</code>	Error function used in the normal CDF
<code>sum(condition)/N</code>	<code>sum(x >= a & x <= b)/N</code>	Empirical probability

3 Exercises

Write all commands yourself in a script or live script. Label figures and answer interpretation questions in comments.

1. Uniform random samples:

- Generate $N = 10000$ samples from $\mathcal{U}[-4, 10]$.
- Display the min and max values of the vector. Why are the endpoints absent? and why does this not change the theoretical distribution?
- Compare `mean()` with $(a + b)/2$.
- Plot the first 100 generated values with `plot()`.
- Plot a 30-bin PDF-normalized histogram. What is its theoretical height?

2. The effect of the uniform-distribution parameters:

- Generate 10000 samples each from $\mathcal{U}[0, 1]$ and $\mathcal{U}[5, 7]$.
- Plot PDF-normalized histograms with equal bin counts in separate windows (`figure`).
- Describe the changes in PDF location, width and height.

3. Normal random samples:

- Set `mu1 = 2`, `sigma1 = sqrt(2)`, and generate 10000 samples from $\mathcal{N}(\mu = 2, \sigma^2 = 2)$ as `x_normal_1`. The second parameter is the variance.
- Plot the first 100 values in the vector.
- Plot the histogram of all values.
- Compute the sample mean and standard deviation with `mean()` and `std()`, and compare with 2 and $\sqrt{2}$. Why are they not exactly equal?
- Keep `sigma1`, change μ to -3 , and compare histograms. What changes?
- Keep `mu1`, change σ to 0.5, and compare again.

4. Probabilities for a uniform variable:

Let $X \sim \mathcal{U}[-4, 10]$.

- Compute theoretically $P(-1 \leq X \leq 3)$.
- Estimate it from Exercise 1 using logical comparisons and `sum()`.
- Compute and estimate $P(X > 7)$.
- Repeat for $N = 100, 1000$, and 100000. How does error tend to change? Must it decrease at every run?

5. Probabilities for a normal variable using `erf()`:

Let $X \sim \mathcal{N}(\mu = 2, \sigma^2 = 2)$.

- a. Use `erf()` to compute the theoretical values of $P(X \leq 3)$, then $P(0 \leq X \leq 4)$, then $P(X > 5)$.
- b. Estimate all three from `x_normal_1`.
- c. Use `normcdf()` function to compute the same probabilities (read the documentation about how to use it).

3.1 Supplementary exercises

6. Write `p = myCDF(v, x)` to estimate $P(V \leq x)$:

$$\hat{F}_V(x) = \frac{\text{number of elements of } v \text{ that are } \leq x}{\text{total number of elements of } v}.$$

Evaluate it at 100 points using a `for` loop. Plot uniform and normal estimates against their theoretical CDFs.

7. Write `p = myPDF(v, x, epsilon)` to estimate the PDF around `x` from samples in $[x - \varepsilon, x + \varepsilon]$:

$$\hat{f}_V(x) = \frac{\text{number of samples in } [x - \varepsilon, x + \varepsilon]}{2\varepsilon N}.$$

Try several ε values. Explain the smoothness-detail trade-off.

4 Final questions

1. Can a PDF be greater than 1? Why can probabilities still not exceed 1?
2. Why can an estimate from $N = 1000$ samples be worse than one from $N = 100$?
3. Two datasets have the same mean and variance. Must they have the same distribution?
4. How could you decide whether measured data are better modelled by a uniform or normal distribution?